

Unit: II
Lecture: 5

Memory Management Issue

According to the size of the object, there are different memory management issues that processor has to take care of:

- For fixed size implementation: For fixed size implementation, like LISP and Unix file system, the only memory management issue is to find the empty space for newly created cell.
- For variable size implementation: For variable size implementation, there are two main memory management issue:
 - Fragmentation
 - Garbage Collection

Garbage Collection: Garbage is the memory which was allocated by the program but is no longer referred. Garbage is the unreachable memory block of allocated areas which cannot be reached directly or indirectly following the pointers. Tuples, arrays, records, strings can all become garbage.

The two major tasks of memory management is to allocate and deallocate the memory according to the requirement. This can be done in two ways:

- Manual Memory Management
- Automatic Garbage Collection

Manual Memory Management: Manual Memory allocation is controlled by the programmer, either implicitly (e.g. by declaring new variables), or explicitly through the issuance of memory allocation commands.

- We need to know explicitly how big of a memory region we want to allocate, and we need to issue an explicit command for the allocation to occur. As a result of successful request we get back a pointer to the newly allocated memory block.
- In a correct program, we need to explicitly deal both with the situation when the memory request succeeds and when it fails.

Advantages of Manual Memory Management:

- Manual memory management is very flexible and it allows the programmer to optimize the sequence of memory allocation/deallocation commands and to adapt these to the semantics of the application at hand.
- It allows programmer to reuse of allocated memory blocks at different stages of the program.

Disadvantages :

- The programmer must deal explicitly with low-level details.
- Errors in manual memory management can occur both at the time of allocation and at the time of deallocation. Some of the errors are:
 - **Memory Leaks:** A memory leak is an unintentional form of memory consumption whereby the programmer fails to free an allocated block of memory when no longer needed.

- **Dangling Pointers:** A dangling pointer is a pointer that points to invalid data or to data which is not valid anymore. When a pointer is pointing at the memory address of a variable but after some time that variable is deleted from that memory location while the pointer is still pointing to it, then such a pointer is known as a dangling pointer.
- **Double free bug:** It is possible to deallocate a memory region more than once. Such events are likely to corrupt the data structures that the memory management system maintains to keep track of allocated and free memory blocks. Such corruption often leads to hard to debug errors and unpredictable program behavior.

Automatic Garbage Collection: It is the process of automatically freeing the space in the heap for allocation of new objects. If garbage collection occurs at unpredictable moments of time, and the program is stopped or significantly slowed down for an extended period of time, then automated garbage collection can not be used in certain critical applications. Program execution should be disrupted as little as possible by garbage collection. Instead of waiting for all memory to be allocated, garbage collectors are invoked periodically.

In this context, the trigger for garbage collection rounds could consist of the issuance, after the previous round of garbage collection:

- of a fixed number of memory allocation requests, or
- of memory allocation requests totaling at least a certain amount of memory, or
- of a number of memory requests proportional to the amount of non-garbage data available at the end of the previous garbage collection round.